

ML Midterm: 100 Practice Items

CS0075 Machine Learning Algorithm. 40 flashcards, 25 import blanks, 25 code blanks, 10 lists. Facts and code come from the midterm reviewer and the M2 slides. The answer key is at the end.

Part A. Flashcards (1 to 40)

1. In the four-stage ML workflow, which stage includes splitting the data?
 - A. Deployment
 - B. Project setup
 - C. Data preparation
 - D. Modeling
2. In the four-stage ML workflow, which stage includes hyperparameter tuning?
 - A. Data preparation
 - B. Project setup
 - C. Deployment
 - D. Modeling
3. Which language ranks number 1 for machine learning in the slides?
 - A. R
 - B. Python
 - C. Julia
 - D. Java
4. Which Python library from the slides is used for web crawling?
 - A. Keras
 - B. SQLAlchemy
 - C. Scrapy
 - D. Seaborn
5. Which Python tool from the slides is a distribution for large-scale data processing and scientific computing?
 - A. Jupyter Notebook
 - B. Anaconda

C. Pandas

D. SciPy

6. What drives a traditional rule-based algorithm?

A. A learned model

B. An accuracy score

C. A condition (true goes to one code, false to another)

D. Training data

7. What is the key difference when machine learning is compared with rule-based programming?

A. The rules are written by hand

B. It only works for simple rules

C. It needs no data

D. The rules are learned automatically from data

8. According to the pie chart, what do data scientists spend most of their time on?

A. Creating training sets

B. Data cleaning and organization (60 percent)

C. Improving algorithms

D. Data mining

9. Which three problems does the slide say real-world data has?

A. Size, speed, cost

B. Bias, variance, noise

C. Incompleteness, noise, inconsistency

D. Missing, scaled, encoded

10. In the messy table, the same Id appearing twice is marked as what?

A. A missing value

B. An attribute dependency

C. A misspelling

D. An invalid duplicate item

11. "Identifies unusual data points" describes which EDA activity?

A. Summary statistics

B. Outlier detection

- C. Correlation analysis
- D. Hypothesis testing

12. Which is an example of discrete data?

- A. Temperature
- B. Height
- C. Weight
- D. Number of students in a class

13. On the Titanic data, which imputation fits the numeric Age column?

- A. Ordinal encoding
- B. Dropping the column
- C. Mean imputation
- D. Most-frequent imputation

14. On the Titanic data, which imputation fits the categorical Embarked column?

- A. Mean imputation
- B. Median imputation
- C. Most-frequent imputation
- D. Standard scaling

15. On the Titanic data, which transformation is applied to Age and Fare?

- A. Ordinal encoding
- B. Feature selection
- C. Most-frequent imputation
- D. Standard scaling

16. Unintended harm to a group, even from good intentions, is called what?

- A. Disparate impact
- B. Data drift
- C. Informed consent
- D. Data portability

17. What does a regression algorithm predict?

- A. Continuous real numbers
- B. Rules

- C. Class labels
- D. Clusters

18. A model that memorizes the training data and does poorly on new data is...

- A. Regularized
- B. Overfitting (high variance)
- C. Underfitting (high bias)
- D. Generalizing well

19. What does a very small k do to a k -NN model?

- A. Makes it linear
- B. Makes it more complex and prone to overfitting
- C. Has no effect
- D. Makes it simpler and underfit

20. What does a very large k do to a k -NN model?

- A. Removes the need for training data
- B. Makes it more complex and overfit
- C. Has no effect
- D. Makes it simpler and prone to underfitting

21. What does `score()` return for a classifier, and for a regressor?

- A. Accuracy for a classifier, R squared for a regressor
- B. Accuracy for both
- C. R squared for both
- D. Mean squared error for both

22. An R squared below zero means the model is...

- A. Perfect
- B. Worse than always predicting the mean
- C. Overfit by a small amount
- D. Exactly as good as the mean

23. In $\hat{y} = w \cdot x + b$, what is b also called?

- A. The learning rate
- B. The slope

- C. The weight vector
- D. The intercept (offset or bias)

24. What does ordinary least squares (OLS) minimize?

- A. The number of features
- B. Mean squared error on the training set
- C. The size of the weights
- D. Absolute error on the test set

25. In Ridge regression, what does a larger alpha do?

- A. Removes the penalty
- B. Applies a bigger penalty and a simpler model
- C. Applies a smaller penalty
- D. Adds more features

26. Which model can set weights to exactly zero?

- A. k-NN
- B. Ridge (L2)
- C. Plain LinearRegression
- D. Lasso (L1)

27. Ridge regression uses which kind of penalty?

- A. L2 regularization
- B. L1 regularization
- C. A tree-depth penalty
- D. No penalty

28. In LogisticRegression, what does a high C mean?

- A. Weaker regularization
- B. Stronger regularization
- C. More neighbors
- D. A smaller tree

29. Despite its name, logistic regression is used for what?

- A. Clustering
- B. Regression

- C. Dimension reduction
- D. Classification

30. Which Naive Bayes class is for continuous features?

- A. BernoulliNB
- B. GaussianNB
- C. MultinomialNB
- D. CategoricalNB

31. Which Naive Bayes class is for integer count data such as word counts?

- A. LinearSVC
- B. GaussianNB
- C. BernoulliNB
- D. MultinomialNB

32. Which of these models needs its features scaled?

- A. SVM
- B. Random forest
- C. Decision tree
- D. Gradient boosting

33. What is a random forest?

- A. A linear model with many features
- B. One very deep tree
- C. An average of many independent trees built on bootstrap samples
- D. Trees built in turn, each fixing the last

34. How does gradient boosting build its trees?

- A. All at once, independently
- B. Without any depth limit
- C. One after another, each correcting the previous trees' errors
- D. From a single bootstrap sample

35. What is the default learning rate of GradientBoostingClassifier?

- A. 1.0
- B. 0.01

C. 0.1

D. 0.5

36. What is a limitation of a decision tree regressor?

A. It needs scaled features

B. It cannot overfit

C. It cannot extrapolate beyond the training data range

D. It only handles two features

37. In an RBF-kernel SVM, what does a small gamma do?

A. Gives a wide radius and a simpler model

B. Turns off the kernel

C. Gives a narrow radius and a more complex model

D. Doubles the number of support vectors

38. In an SVM, what does a small C do?

A. Switches to a linear kernel

B. Lets every point influence it freely

C. Scales the features

D. Restricts the model heavily

39. Which is a weakness of SVMs?

A. Slow with many samples and sensitive to feature scaling

B. They cannot do classification

C. They only fit linear data

D. They need no parameter tuning

40. On the breast cancer data, the default SVC on unscaled features scores train 1.00 and test 0.63. What does that show?

A. Severe overfitting caused by unscaled features

B. A perfect model

C. Too few samples

D. Underfitting

Part B. Fill In The Blank: Imports (41 to 65)

41. import ____ as np
42. import pandas as ____
43. import matplotlib.____ as plt
44. import ____ (the textbook's plotting and dataset helper package)
45. from sklearn.____ import load_iris
46. from sklearn.datasets import ____ (the 569-sample cancer data)
47. from sklearn.datasets import ____ (the two_moons data)
48. from sklearn.model_selection import ____
49. from sklearn.____ import KNeighborsClassifier
50. from sklearn.neighbors import ____ (k-NN for a continuous target)
51. from sklearn.impute import ____
52. from sklearn.preprocessing import ____ (z-score scaling)
53. from sklearn.preprocessing import ____ (scales to 0 to 1)
54. from sklearn.____ import accuracy_score
55. from sklearn.linear_model import ____ (ordinary least squares)
56. from sklearn.linear_model import ____ (L2 penalty)
57. from sklearn.linear_model import ____ (L1 penalty, can zero out weights)
58. from sklearn.linear_model import ____ (a classifier despite its name)
59. from sklearn.____ import SVC
60. from sklearn.svm import ____ (the linear SVM)
61. from sklearn.____ import DecisionTreeClassifier
62. from sklearn.tree import ____ (writes a .dot file)
63. import ____ (shows the .dot file inside Python)
64. from sklearn.ensemble import ____ (many independent trees)
65. from sklearn.____ import GradientBoostingClassifier

Part C. Fill In The Blank: Code (66 to 90)

66. X_train, X_test, y_train, y_test = train_test_split(X, y, ____=0)

67. `train_test_split(cancer.data, cancer.target, ____=cancer.target, random_state=66)` (keeps class proportions)
68. `knn = KNeighborsClassifier(____=3)`
69. `knn.____(X_train, y_train)` (trains the model)
70. `prediction = knn.____(X_test)`
71. `print("Test set score: {:.2f}".format(knn.____(X_test, y_test)))`
72. `X_new = np.____([[40, 80, 70, 30]])`
73. `reg = ____ (n_neighbors=3)` (k-NN for a continuous target)
74. `lr = LinearRegression().____(X_train, y_train)`
75. `lr.coef_` holds the weights. Which attribute holds the intercept? `lr.____`
76. `lasso001 = Lasso(____=0.01, max_iter=100000).fit(X_train, y_train)`
77. `lasso001 = Lasso(alpha=0.01, ____=100000).fit(X_train, y_train)`
78. `logreg100 = LogisticRegression(____=100).fit(X_train, y_train)`
79. `lr_l1 = LogisticRegression(C=C, penalty="____", solver='liblinear')` (the L1 penalty)
80. `lr_l1 = LogisticRegression(C=C, penalty="l1", solver='____')`
81. `tree = DecisionTreeClassifier(____=4, random_state=0)` (pre-pruning)
82. `forest = RandomForestClassifier(____=100, random_state=0)`
83. `gbdt = GradientBoostingClassifier(random_state=0, ____=0.01)`
84. `svm = SVC(kernel='____', C=10, gamma=0.1).fit(X, y)`
85. `svc = SVC(____=1000)` (the improvement on scaled data)
86. `min_on_training = X_train.min(____=0)`
87. `range_on_training = (X_train - min_on_training).____(axis=0)`
88. `X_test_scaled = (X_test - ____) / range_on_training` (reuse the TRAINING values)
89. `imputer = SimpleImputer(strategy='____', missing_values=np.nan)` (fill with the average)
90. `df['Age'] = imputer.____(df[['Age']]).ravel()`

Part D. Lists (91 to 100)

91. Name the four stages of the ML workflow. (4 items)
92. Name the four types of machine learning shown in the picture. (4 items)
93. Name the three data quality problems. (3 items)

- 94.** Name the five data ethics principles. (5 items)
- 95.** Name the six key steps of data preprocessing. (6 items)
- 96.** What do E, T and P stand for in the formal definition of ML? (3 items)
- 97.** Name the three Naive Bayes classes in scikit-learn. (3 items)
- 98.** `train_test_split` returns four items. Name them. (4 items)
- 99.** Name the three linear regression models covered in the slides. (3 items)
- 100.** Name the two main parameters of an RBF-kernel SVM. (2 items)

Answer Key

No.	Answer	Full line (blanks only)
1	Data preparation	
2	Modeling	
3	Python	
4	Scrapy	
5	Anaconda	
6	A condition (true goes to one code, false to another)	
7	The rules are learned automatically from data	
8	Data cleaning and organization (60 percent)	
9	Incompleteness, noise, inconsistency	
10	An invalid duplicate item	
11	Outlier detection	
12	Number of students in a class	
13	Mean imputation	
14	Most-frequent imputation	
15	Standard scaling	
16	Disparate impact	
17	Continuous real numbers	
18	Overfitting (high variance)	
19	Makes it more complex and prone to overfitting	
20	Makes it simpler and prone to underfitting	
21	Accuracy for a classifier, R squared for a regressor	
22	Worse than always predicting the mean	
23	The intercept (offset or bias)	
24	Mean squared error on the training set	
25	Applies a bigger penalty and a simpler model	
26	Lasso (L1)	
27	L2 regularization	
28	Weaker regularization	
29	Classification	
30	GaussianNB	
31	MultinomialNB	
32	SVM	
33	An average of many independent	

	trees built on bootstrap samples	
34	One after another, each correcting the previous trees' errors	
35	0.1	
36	It cannot extrapolate beyond the training data range	
37	Gives a wide radius and a simpler model	
38	Restricts the model heavily	
39	Slow with many samples and sensitive to feature scaling	
40	Severe overfitting caused by unscaled features	
41	numpy	Full line: import numpy as np
42	pd	Full line: import pandas as pd
43	pyplot	Full line: import matplotlib.pyplot as plt
44	mglearn	Full line: import mglearn
45	datasets	Full line: from sklearn.datasets import load_iris
46	load_breast_cancer	Full line: from sklearn.datasets import load_breast_cancer
47	make_moons	Full line: from sklearn.datasets import make_moons
48	train_test_split	Full line: from sklearn.model_selection import train_test_split
49	neighbors	Full line: from sklearn.neighbors import KNeighborsClassifier
50	KNeighborsRegressor	Full line: from sklearn.neighbors import KNeighborsRegressor
51	SimpleImputer	Full line: from sklearn.impute import SimpleImputer
52	StandardScaler	Full line: from sklearn.preprocessing import StandardScaler
53	MinMaxScaler	Full line: from sklearn.preprocessing import MinMaxScaler
54	metrics	Full line: from sklearn.metrics import accuracy_score
55	LinearRegression	Full line: from sklearn.linear_model import LinearRegression
56	Ridge	Full line: from sklearn.linear_model import Ridge
57	Lasso	Full line: from sklearn.linear_model import Lasso
58	LogisticRegression	Full line: from sklearn.linear_model import LogisticRegression
59	svm	Full line: from sklearn.svm import SVC
60	LinearSVC	Full line: from sklearn.svm import LinearSVC
61	tree	Full line: from sklearn.tree import

		DecisionTreeClassifier
62	export_graphviz	Full line: from sklearn.tree import export_graphviz
63	graphviz	Full line: import graphviz
64	RandomForestClassifier	Full line: from sklearn.ensemble import RandomForestClassifier
65	ensemble	Full line: from sklearn.ensemble import GradientBoostingClassifier
66	random_state	Full code: train_test_split(X, y, random_state=0)
67	stratify	Full code: train_test_split(cancer.data, cancer.target, stratify=cancer.target, random_state=66)
68	n_neighbors	Full code: knn = KNeighborsClassifier(n_neighbors=3)
69	fit	Full code: knn.fit(X_train, y_train)
70	predict	Full code: prediction = knn.predict(X_test)
71	score	Full code: knn.score(X_test, y_test)
72	array	Full code: X_new = np.array([[40, 80, 70, 30]]) (2-D, so a list inside a list)
73	KNeighborsRegressor	Full code: reg = KNeighborsRegressor(n_neighbors=3)
74	fit	Full code: lr = LinearRegression().fit(X_train, y_train)
75	intercept_	Full code: lr.intercept_
76	alpha	Full code: Lasso(alpha=0.01, max_iter=100000)
77	max_iter	Full code: Lasso(alpha=0.01, max_iter=100000)
78	C	Full code: LogisticRegression(C=100)
79	l1	Full code: LogisticRegression(C=C, penalty="l1", solver='liblinear')
80	liblinear	Full code: LogisticRegression(C=C, penalty="l1", solver='liblinear')
81	max_depth	Full code: DecisionTreeClassifier(max_depth=4, random_state=0)
82	n_estimators	Full code: RandomForestClassifier(n_estimators=100, random_state=0)
83	learning_rate	Full code: GradientBoostingClassifier(random_state=0, learning_rate=0.01)
84	rbf	Full code: SVC(kernel='rbf', C=10, gamma=0.1)
85	C	Full code: svc = SVC(C=1000)
86	axis	Full code: min_on_training = X_train.min(axis=0)
87	max	Full code: range_on_training = (X_train - min_on_training).max(axis=0)

88	min_on_training	Full code: <code>X_test_scaled = (X_test - min_on_training) / range_on_training</code>
89	mean	Full code: <code>SimpleImputer(strategy='mean', missing_values=np.nan)</code>
90	fit_transform	Full code: <code>df['Age'] = imputer.fit_transform(df[['Age']]).ravel()</code>
91	Project setup, Data preparation, Modeling, Deployment	
92	Supervised, Unsupervised, Semi-supervised, Reinforcement	
93	Incompleteness, Noise, Inconsistency	
94	Ownership, Transparency, Privacy, Intention, Outcomes	
95	Profiling, Cleansing, Reduction, Transformation, Enrichment, Validation	
96	Experience, Task, Performance	
97	GaussianNB, BernoulliNB, MultinomialNB	
98	X_train, X_test, y_train, y_test	
99	Linear Regression, Ridge, Lasso	
100	C, gamma	